

Technology Stack

Date	02 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 Templates	Provides a responsive and user-friendly interface. Jinja2 dynamically displays HDI prediction results without requiring a JavaScript framework.
2	Backend / Server-Side	Python 3.11, Flask 3.0.3	Flask is lightweight, easy to develop, and integrates seamlessly with machine learning models for prediction.
3	Database / Data Storage	CSV Dataset (HDI.csv)	Stores HDI and socioeconomic data in a simple format suitable for machine learning training and analysis.
4	Cloud / Hosting / Deployment	Local Flask Server / VS Code	Enables easy local deployment, testing, and demonstration without requiring cloud infrastructure.
5	Version Control & CI/CD	Git, GitHub	Tracks source code changes, supports collaboration, and maintains project version history.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Pickle, Jupyter Notebook	Scikit-learn trains the Linear Regression model, Pandas and NumPy handle data processing, Matplotlib and Seaborn create visualizations, Pickle saves the trained model, and Jupyter Notebook is used for EDA and model development.